# RESERVOIR SIMULATOR

3D Reservoir Modeling & Production Optimization

DATABRICKS APPS

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## INTERACTIVE 3D RESERVOIR SIMULATION WITH AI-POWERED FIELD DEVELOPMENT PLANNING

Reservoir Simulator delivers an interactive 3D reservoir model with scenario management, well performance analysis, economic evaluation, and SAP ERP integration — calibrated to the Norne field (North Sea) benchmark dataset.

### ◆ BUSINESS VALUE

- **Optimize $100M+ capital allocation** with scenario-based field development planning and NPV analysis
- **Visualize reservoir behavior in 3D** with oil/water saturation heatmaps, well trajectories, and layer navigation
- **Compare development scenarios** side-by-side with production, economics, and efficiency delta analysis
- **Integrate cost tracking** with SAP BDC for CAPEX/OPEX, procurement, and vendor contract management
- **Accelerate reservoir analysis** with Foundation Model API-powered AI assistant for optimization

### ◆ CHALLENGES ADDRESSED

- **Expensive simulation software** — high-cost commercial tools with steep learning curves
- **Limited scenario comparison** — no easy way to compare multiple development plans side-by-side
- **Disconnected economics** — reservoir models and financial analysis in separate tools
- **No AI integration** — manual interpretation of simulation results without automated recommendations
- **Poor collaboration** — desktop-only tools preventing web-based team review of models

### ◆ WHO IT'S FOR

🔬 **Reservoir Engineer**
3D visualization, scenario management, well results, and pressure tracking

💰 **Economics/Planning Analyst**
NPV, IRR, payout period, cashflow analysis, and fiscal term modeling

📊 **Operations Manager**
Field-level production summaries, well status, and operational KPIs

🤖 **Development Geologist**
AI-powered reservoir analysis and well placement optimization

| 5 | 1,000 | 9 | 40 |
|---|---|---|---|
| Wells | Grid Cells | Dashboard Tabs | Timesteps |

### ◆ QUICK START GUIDE

1. **Clone Repository**
   Clone the Reservoir Simulator repo to your local machine
2. **Build Frontend**
   Run cd frontend && npm install && npm run build
3. **Deploy the App**
   Run databricks apps deploy to launch the simulator
4. **Create Scenarios**
   Configure wells, injection, and economic assumptions

### ◆ GETTING THE MOST OUT OF RESERVOIR SIMULATOR

🌐 **Explore 3D Reservoir**
Navigate the Three.js viewport with saturation heatmaps, well paths, and layer views

📊 **Compare Scenarios**
Run side-by-side comparison of multiple simulation runs with deltas

💰 **Analyze Economics**
Configure price decks and fiscal terms for NPV, IRR, and cashflow analysis

🤖 **Consult Reservoir Agent**
Chat with the AI assistant for reservoir analysis and optimization

### ◆ TARGET INDUSTRY

**Upstream Oil & Gas** — Reservoir engineering and field development planning for conventional and unconventional reservoirs. Calibrated to the Norne field (North Sea) benchmark.

⚠ **DEMO ONLY** — This application is a demonstration Databricks App. Please refer to the included LICENSE file for full terms of use.

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